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RESEARCH

Graphene Metasurface-Based Surface Plasmon Resonance Biosensor for Rapid COVID-19 Detection with Machine Learning Optimization

Jacob Wekalao[1]

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Abstract
This study presents a graphene metasurface-based surface plasmon resonance biosensor for rapid and sensitive detection of SARS-CoV-2. The sensor design incorporates four identical circular resonators coated with graphene monolayer on a SiO_2 substrate, operating in the terahertz frequency range of 0.6–1.4 THz. The sensor demonstrates a quality factor of 7.028 and sensitivity of 400 $GHzRIU^{-1}$, with optimal performance observed at 1.002–1.004 THz. Electromagnetic simulations using COMSOL Multiphysics exemplifies the sensor's response to variations in graphene chemical potential, angle of incidence, and resonator dimensions. The detection capability is enhanced through machine learning optimization utilizing 1D Convolutional Neural Networks, achieving perfect R^2 scores of 100% under specific operational parameters. The proposed sensor offers a promising platform for rapid, sensitive, and reliable COVID-19 detection, particularly suitable for point-of-care applications in resource-limited settings.

Keywords Biosensor · Terahertz sensing · Point-of-care diagnostics · SARS-CoV-2 · Electromagnetic simulation · Neural networks

Introduction

The emergence of SARS-CoV-2 has precipitated an unprecedented global health crisis, with the virus's ability to rapidly mutate creating ongoing challenges for healthcare systems worldwide[1]. This pandemic has highlighted the critical need for portable, scalable, affordable, and compact solutions for point-of-care (PoC) viral detection, particularly in resource-limited settings[2]. The ability to quickly identify infected individuals is paramount to controlling transmission and effectively managing outbreaks, making the development of accessible diagnostic tools an urgent priority in the collective response to COVID-19 and future pandemic threats[3].

SARS-CoV-2 is distinguished by its complex structure, featuring prominent spike glycoproteins that give the virus its characteristic crown-like appearance, a nucleocapsid protein that protects its RNA genome, and an outer lipid bilayer derived from host cell membranes[4]. Unlike its coronavirus predecessors SARS-CoV and MERS-CoV, SARS-CoV-2 demonstrates enhanced binding affinity to human ACE2 receptors through its spike proteins, facilitating more efficient cell entry and contributing to its heightened transmissibility[5]. These spike proteins represent crucial targets for diagnostic assays, as their detection can directly indicate the presence of viable virus particles in clinical samples [6].

While real-time RT-PCR remains the gold standard for COVID-19 diagnosis due to its exceptional sensitivity and specificity in detecting viral RNA, its implementation presents significant challenges in pandemic response[7]. The method requires advanced laboratory infrastructure, expensive thermal cycling equipment, cold chain maintenance, and highly trained personnel—resources often unavailable in low-resource settings or during surge capacity needs[8]. These limitations have spurred interest in alternative nucleic acid amplification techniques such as LAMP (loop-mediated isothermal amplification), which offers advantages including isothermal processing, faster turnaround times, and simplified equipment requirements[9]. However, LAMP faces its own implementation challenges, including optimization difficulties, potential for contamination, and the need for careful primer design to maintain specificity while accommodating viral mutations[10].

✉ Jacob Wekalao
jacob1902@mail.ustc.edu.cn

1 Department of Optics and Optical Engineering, University of Science and Technology of China, Hefei 230026, China

Biosensor technology has emerged as a revolutionary approach for viral pathogen detection, offering rapid, sensitive, and label-free detection capabilities that overcome many limitations of conventional diagnostic methods[11]. These devices integrate biological recognition elements with physicochemical transducers to convert specific biomolecular interactions into measurable signals, enabling direct detection of viral components without the need for amplification steps or complex sample preparation[12]. The simplicity and speed of biosensing platforms make them valuable for point-of-care applications where timely diagnosis is critical for effective disease management and containment[13].

Surface plasmon resonance occurs when p-polarized light strikes a thin metal film at a specific angle, creating evanescent waves that interact with free electrons in the metal, resulting in collective electron oscillations known as surface plasmons[14]. This resonance condition is highly dependent on the refractive index of the medium adjacent to the metal film[15]. When biomolecules bind to recognition elements on the sensor surface, they alter the local refractive index, shifting the resonance conditions in a manner proportional to the mass of bound analyte[16]. This shift can be measured by monitoring changes in the angle, wavelength, or intensity of the reflected light, providing quantitative information about binding events in real-time[17].

Traditional SPR sensors predominantly utilize noble metals, with gold (Au) and silver (Ag) being the most common materials for the plasmonic sensing layer[18]. Gold offers excellent chemical stability, biocompatibility, and well-established surface chemistry for biomolecule immobilization, making it the preferred choice for most biosensing applications[19]. However, gold exhibits relatively broad SPR curves, which can limit detection accuracy for small refractive index changes [20–22]. Silver, conversely, produces sharper SPR curves and higher sensitivity due to its superior plasmonic properties, but suffers from poor chemical stability and susceptibility to oxidation, limiting its practical utility in biological environments [23–27].

Researchers have made tremendous efforts in exploring biosensing technologies. For instance, Vasanthi et al. presented a boron nitride (BN)-integrated SPR biosensor for early colorectal cancer detection, achieving a sensitivity of 272.27°/RIU, a quality factor of 64.024 RIU‾1, and a detection accuracy of 0.7683[28]. Salah et al. presented a fiber-optical SPR sensor for Helicobacter pylori detection, achieving a maximum sensitivity of 6812.50 nm/RIU, a quality factor (QF) of 84.507 RIU‾1, and a limit of detection (LoD) in RIU[29]. Bouandas et al. developed a high-sensitivity SPR sensor incorporating BK7, copper, TiO_2, nickel, and black phosphorus (BP), achieving a sensitivity of 518°/RIU and a quality factor (QF) of 91.51/RIU with optimized layer thicknesses. An alternative configuration further improved sensitivity to 526°/RIU and QF to 96.51/RIU[30]. Al hawari et al. developed an SPR biosensor for uric acid detection, combining BlueP, Ag, ITO, graphene, and TMDCs, achieving a figure of merit (FOM) of 2638.29 RIU‾1 after optimizing layer thicknesses. This high FOM highlights the sensor's potential for advanced biosensing applications[31]. Shivangan et al. designed a high-performance SPR sensor using perovskite material PbTiO3 and silver for early-stage malaria detection, achieving a sensitivity of 410.8°/RIU and a quality factor of 105.18 RIU‾1. The sensor demonstrated a detection accuracy of 0.21 deg‾1, showcasing its potential for precise malaria diagnosis[32].

This study introduces a sensor based on graphene metasurface technology. Section 2 presents the design methodology and modelling of the metasurface-based sensor, including the graphene conductivity model and key performance parameters. The significant findings regarding transmittance, how structural parameters influence transmittance response, and sensitivity analysis—incorporating sensitivity, uncertainty, and sensor resolution metrics—are examined in Sect. 3. Section 4 covers machine learning optimization techniques, with concluding remarks provided in Sect. 5.

Geometry and Materials

The proposed sensor design consists of four identical circular resonators, each with a radius of 3 µm, arranged on a SiO_2 substrate. The resonators are coated with a monolayer of graphene, which plays a pivotal role in enhancing the sensor's sensitivity to changes in refractive index. The resonators are fabricated on the SiO_2 substrate, which has dimensions of 20 µm by 20 µm in the horizontal plane and a thickness of 1.5 µm, providing structural support and a stable environment for the resonators. This sensor, based on advanced metasurface technology, was simulated using COMSOL Multiphysics software, leveraging the Finite Element Method (FEM) for detailed electromagnetic simulations. The simulation begins with the SiO_2 substrate, onto which the graphene-coated circular resonators are placed. Periodic boundary conditions were implemented along both the x and y axes during the simulation to model a periodic structure effectively, allowing for accurate analysis of the sensor's performance across a wide range of operating conditions. The meshing of the simulation domain was performed using Delaunay tetrahedral tessellation, ensuring a highly refined mesh to capture the electromagnetic behaviour at the interface between the graphene and the surrounding medium. Figure 1 a-d provide detailed visual representations of the metasurface sensor design. Figure 1a presents the top view of the sensor layout, while Fig. 1b shows the 3D view for better understanding of the spatial arrangement of the resonators. Figure 1c illustrates the horizontal view, providing insight into the sensor's cross-sectional structure, and Fig. 1d highlights

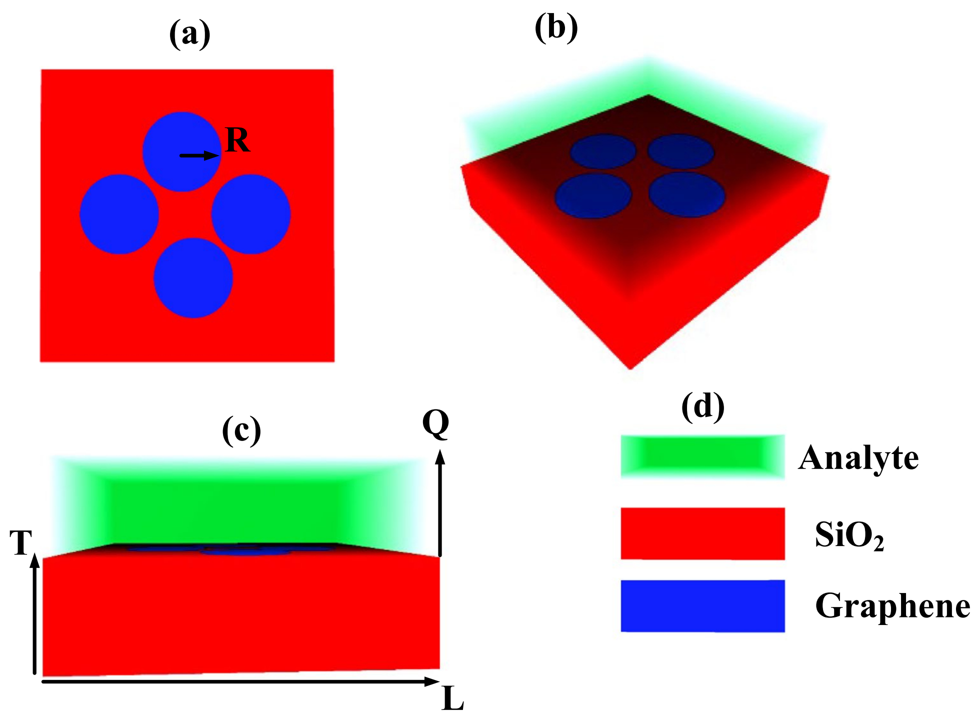

Fig. 1 Schematic representation of the proposed sensor, depicting (**a**) top-down view, **b** three-dimensional perspective, **c** frontal elevation, and (**d**) material composition

the materials used in the sensor design, including graphene and SiO_2, showcasing their arrangement on the metasurface.

Graphene Model

The conductivity σs of graphene is calculated using Eqs. 2 through Γ 3 [33–42]

$$\varepsilon(\omega) = 1 + \frac{\sigma_s}{\varepsilon_0 \omega \nabla} \tag{1}$$

$$\sigma_{intra} = \frac{-je^2 k_B T}{\pi h^2(\omega - j2\tau)} \left(\frac{\mu_c}{k_B T} + 2\ln\left(\frac{\mu_c}{e^k B^T + 1} \right) \right) \tag{2}$$

$$\sigma_{inetr} = \frac{-je^2}{4\pi h} \ln\left(\frac{2\left|\mu_c\right| - (\omega - j2\tau)h}{2\left|\mu_c\right| + (\omega - j2\tau)h} \right) \tag{3}$$

$$\mu_c = E_f \approx hv_f = \sqrt{\frac{\pi \varepsilon_o \varepsilon_o v_g}{ed_{ox}}} \tag{4}$$

Table 1 Various mutated strains of COVID-19 [57]

WHO label	Pango lineage	GISAID clade	Nextstrain clade	Additional amino acid changes monitored	Earliest documented samples	Date of designation
Alpha	B.1.1.7	GRY	20I (V1)	+S:484 K +S:452R	United Kingdom, Sep 2020	Dec 18, 2020
Beta	B.1.351	GH/501Y.V2	20H (V2)	+S:L18F	South Africa, May 2020	Dec 18, 2020
Gamma	P.1	GR/501Y.V3	20 J (V3)	+S:681H	Brazil, Nov 2020	Jan 11, 2021
Delta	B.1.617.2	G/478 K.V1	21A, 21I, 21 J	+S:417N +S:484 K	India, Oct 2021	VOI: Apr 4, 2021 VOC: May 11, 2021
Lambda	C.37	GR/452Q.V1	21G	-	Peru, Dec 2020	Jun 14, 2021
Mu	B.1.621	GH	21H	-	Colombia, Jan 2021	Aug 30, 2021
Omicron	B.1.1.529	GRA	21 K, 21L, 21 M	+R346K	Multiple countries, Nov 2021	VUM: Nov 24, 2021 VOC: Nov 26, 2021

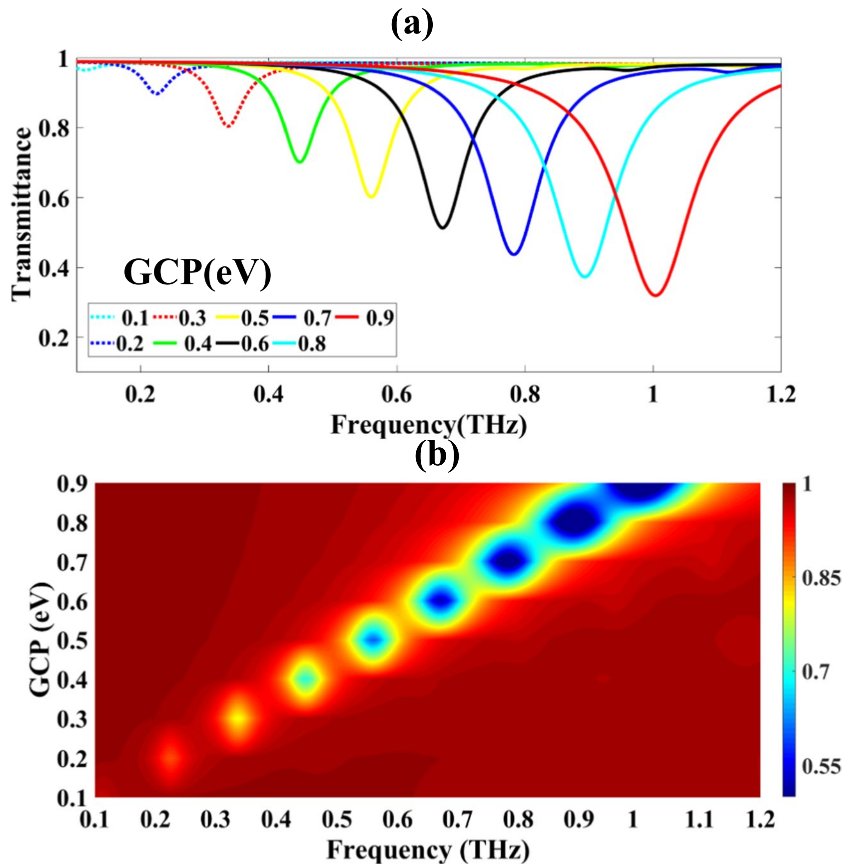

Fig. 2 Shows the relationship between changes in graphene's chemical potential and the resulting transmission response of the sensor

Fabrication of the Proposed Sensor Design

To fabricate the proposed sensor, a series of advanced microfabrication and materials deposition steps are required. The process begins with preparing the SiO_2 substrate, which serves as the foundational material for the sensor. The SiO_2 substrate is chosen for its mechanical stability and compatibility with graphene. The substrate is first thoroughly cleaned using a combination of solvent cleaning (acetone and isopropanol) and an oxygen plasma treatment, ensuring a clean and hydrophilic surface for subsequent layers. Next, the graphene is synthesized using Chemical Vapor Deposition (CVD) on a copper foil, where large-area monolayer graphene sheets are grown. Once the graphene has been produced, it is transferred from the copper foil to the SiO2 substrate. This is achieved through a polymer-assisted transfer method, where the graphene is coated with a layer of poly (methyl methacrylate) (PMMA). The copper foil is then etched away, leaving behind the graphene sheet, which is transferred to the SiO_2 substrate. After transfer, the PMMA layer is removed using acetone or other solvents. The graphene's quality and monolayer nature are then confirmed through characterization techniques such as Raman spectroscopy and Atomic Force Microscopy (AFM). Once the graphene is in place, the circular resonators are defined using electron beam lithography (e-beam lithography). This technique allows for precise nanoscale patterning on the graphene surface. A thin layer of electron-sensitive resist (e.g., PMMA) is first spin-coated onto the graphene-covered SiO_2 substrate. The resist is then exposed to a focused electron beam, which writes the pattern of the resonators onto the resist layer. After exposure, the resist is developed in a suitable developer solution, leaving behind the patterned areas that define the resonator shapes. To etch away the graphene and reveal the resonator patterns, reactive ion etching (RIE) is employed. The RIE process uses an oxygen or argon plasma to selectively remove the graphene in the areas where the resist layer has been developed. This ensures that

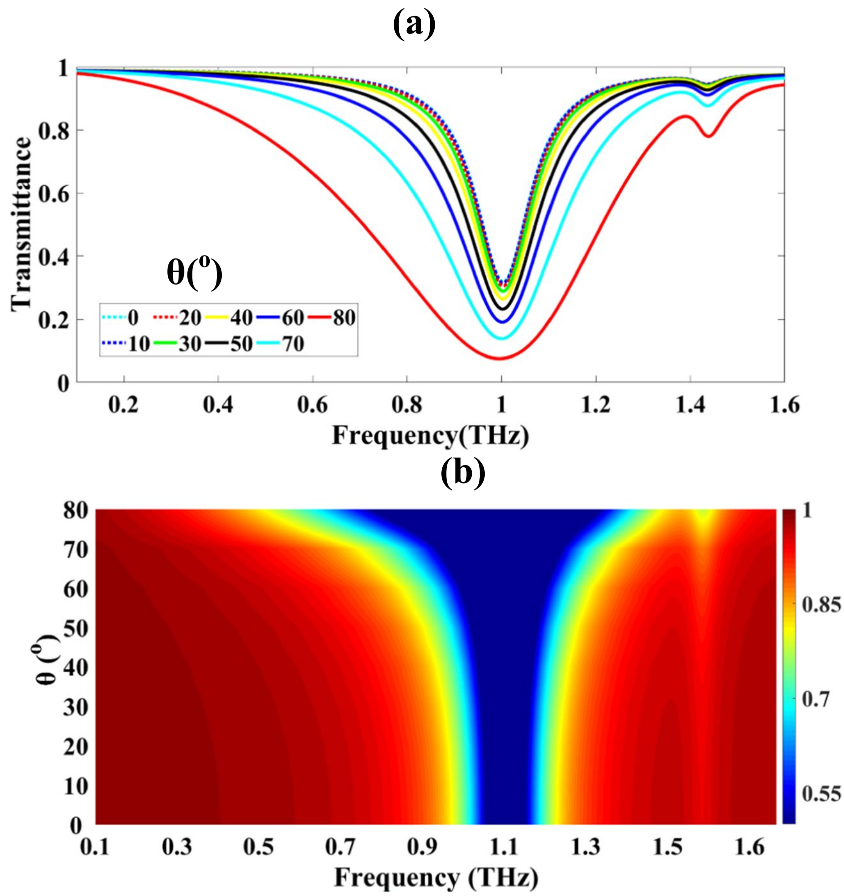

Fig. 3 Shows the relationship between changes in the angle of incidence and the resulting transmission response of the sensor

only the defined resonator structures remain intact, with the rest of the graphene etched away. The SiO_2 substrate, which is dimensioned to 20 µm by 20 µm and 1.5 µm thick, can be precisely cut and polished using microtome or diamond blade techniques to meet the design specifications. Once the graphene resonators are in place, a thin metal layer, such as gold or silver, can be deposited onto the resonators to enhance their sensitivity. This is done using thermal evaporation or sputter deposition, where metal is deposited in a controlled manner onto the surface. Further patterning steps may be required to define additional structural features of the metasurface. These steps might involve additional electron beam lithography or photolithography, depending on the design requirements. The final metasurface, consisting of the graphene-coated resonators, is then integrated with the overall sensor device, ensuring precise positioning of the resonators on the SiO2 substrate. Once the fabrication process is complete, the sensor undergoes extensive testing and characterization. Techniques such as scanning electron microscopy (SEM), atomic force microscopy (AFM), and Raman spectroscopy are used to verify the quality of the graphene, the accuracy of the resonator patterns, and the overall performance of the metasurface. The sensor's sensitivity to refractive index changes is also tested in controlled environments using various liquids or gases to simulate chemical or biological sensing applications. Finally, the sensor is encapsulated in suitable packaging to protect it and facilitate integration into a measurement system. The sensor can be connected to external equipment for real-time monitoring and data collection. If necessary, the individual sensors can be integrated into a sensor array for multiplexed sensing applications. In real-time use, the sensor can be exposed to different analytes, such as blood or chemical vapours, and its response can be monitored to evaluate its performance for applications like chemical detection, biosensing, or environmental monitoring.

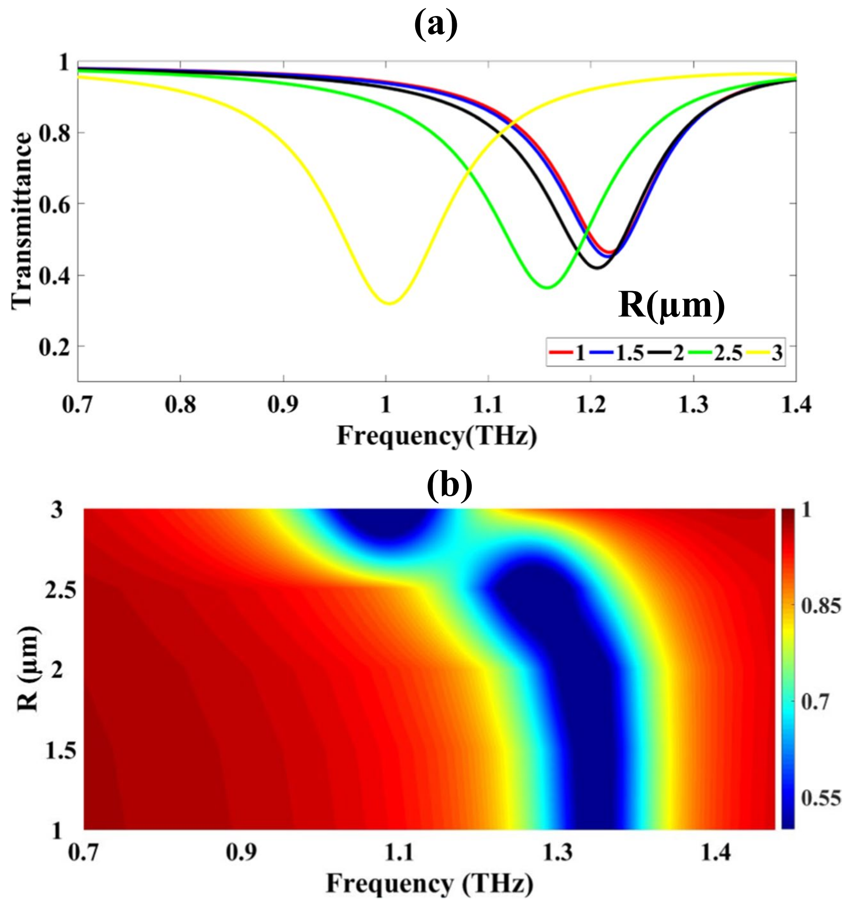

Fig. 4 Shows the relationship between changes in the circular resonator and the resulting transmission response of the sensor

Metasurface Analysis

The electric field component of light is defined within periodic boundaries as[43]:

$$\mathrm{E} = e^{-jk_F.r} \tag{5}$$

The x and y components of the vector K_F are defined by[44]:

$$K_F = \left\{ K_{F_x}, K_{F_y} \right\} = \left\{ \frac{2\pi}{\lambda} \sqrt{\varepsilon} cos\theta, \frac{2\pi}{\lambda} \sqrt{\varepsilon} cos\theta \right\} \tag{6}$$

$$H_{TM} = \left\{ 0, e^{-j(\frac{2\pi}{\lambda} \sqrt{\varepsilon_s} cos\theta.x + \frac{2\pi}{\lambda} \sqrt{\varepsilon_s} sin\theta.z}, 0 \right\} \tag{7}$$

he proposed sensor's impedance is characterized by[45–56];

$$z = \pm \sqrt{\frac{\left(1 + s_{11}\right)^2 - s_{21}^2}{\left(1 - s_{11}\right)^2 - s_{21}^2}} \tag{8}$$

$$e^{ink_0 d} = \frac{s_{11}}{1 - s_{11} \frac{\text{<spanclass='convertEndash'>2−1}}{2+1}} \tag{9}$$

$$n = \frac{1}{k_0 d} \{ [[lne^{ink_0 d})]'' + 2m\pi \} - i[ln(e^{ink_0 d})]'] \tag{10}$$

$$\varepsilon = \frac{n}{z} \tag{11}$$

$$\mu = nz \tag{12}$$

$$A(w) = 1 - R(w) - T(w) = 1 - \left|s_{11}\right|^2 - \left|s_{21}\right|^2 \tag{13}$$

Table 1 provides an overview of various mutated strains of COVID-19, detailing their WHO labels, Pango lineages, and key amino acid changes. It also includes the earliest documented samples and dates of designation for each strain, highlighting their global emergence and classification.

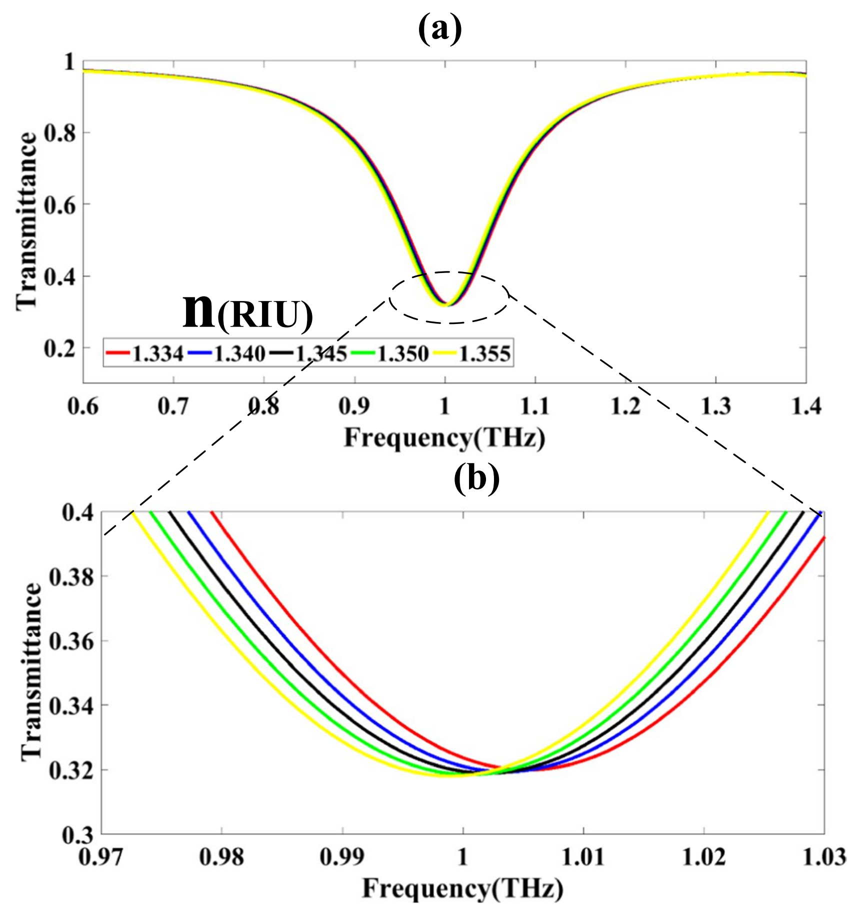

Fig. 5 illustrates how the transmission patterns shift in response to refractive index changes when detecting Covid-19

Results and Discussion

The proposed sensor design presented in Fig. 1 is simulated using COMSOL Multiphysics. The simulation process involves varying multiple fundamental and structural parameters, such as the material composition, dimensions of the sensing element, and the arrangement of the optical components. These variations allow for the identification of the optimal sensor configuration that maximizes performance. The transmittance response of the system is recorded as the key output, which serves to evaluate the sensor's efficiency and sensitivity in detecting specific biomolecules.

The sensor's electromagnetic behaviour was first examined by studying how the Graphene Chemical Potential (GCP) affects its transmission properties. In this analysis, the GCP was modulated from 0.1 eV to 0.9 eV in regular steps of 0.1 eV as depicted in Fig. 2a-b. This range was chosen to observe the sensor's response across a broad spectrum of chemical potential values. The transmittance values obtained for each GCP step are as follows: 96.415%, 89.674%, 80.315%, 70.065%, 60.183%, 51.322%, 43.699%, 37.281%, and 31.939%. These values indicate a significant decrease in transmittance as the GCP increases, which correlates with changes in the material's electronic properties. Additionally, the color plot in Fig. 2b clearly demonstrates a rightward shift in the transmittance response, reflecting the progressive reduction in transmittance as GCP is increased. This shift is indicative of the sensor's sensitivity to changes in the electronic environment of the graphene layer.

The sensor's electromagnetic behaviour was further investigated by studying how the angle of incidence affects its transmission properties. In this study, the angle of incidence was modulated from 0° to 80° in regular steps of 10° as depicted in Fig. 3a-b. The transmittance values obtained at each step are as follows: 31.939%, 31.613%, 30.625%, 28.942%, 26.514%, 23.267%, 19.099%, 13.898%, and 7.535%. These results show a progressive decrease in transmittance as the angle of incidence increases, which indicates a reduction in the sensor's ability to transmit light as the angle deviates from normal. Additionally, the color plot in Fig. 3b illustrates a widening of the transmittance band with an increase in the angle of incidence. This widening effect suggests that the sensor's sensitivity to angular variation is significant

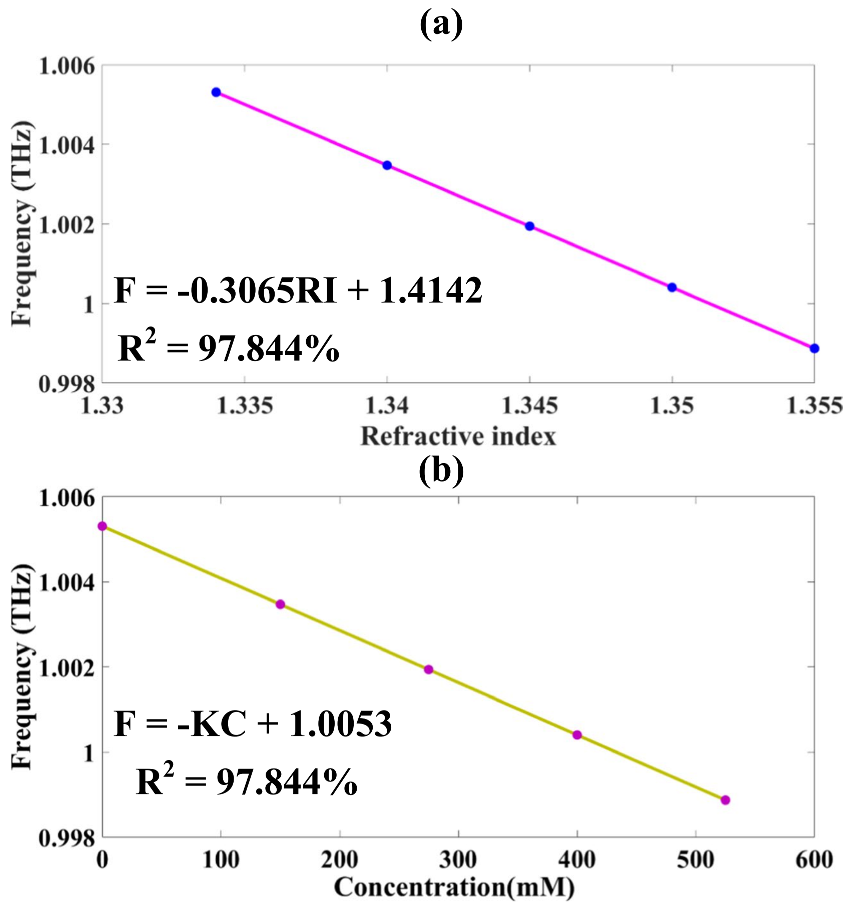

Fig. 6 **a-b** Graphs showing how resonant frequency correlates with refractive indices and various concentrations in Covid-19 samples

and that it can effectively detect changes in the incident light's angle.

The sensor's electromagnetic behaviour was examined by studying how the variation in the dimensions of circular resonator designs affects its transmission properties. In this analysis, the dimensions of the circular resonators were modulated from 1 µm to 3 µm, with a step size of 0.5 µm as depicted in Fig. 4a-b. The transmittance values observed for each resonator dimension are as follows: 46.279%, 45.082%, 41.881%, 36.390%, 31.938%, and 7.535%. These results indicate a significant drop in transmittance as the resonator size increases, which can be attributed to changes in the electromagnetic field interactions within the resonator structure. Additionally, the color plot in Fig. 4b clearly demonstrates a leftward shift in the transmittance response as the dimensions of the circular resonators increase. This shift suggests that larger resonators tend to absorb and scatter more light, reducing the sensor's overall transmittance.

Detection Analysis

After obtaining the optimized structure of the proposed sensor, we applied it to the detection of COVID-19. The sensor's response, shown in Fig. 5a and b, illustrates the performance under specific conditions. In Fig. 5a, we observe the transmittance drops across various frequencies, which are recorded as 31.990%, 31.939%, 31.893%, 31.853%, and 31.807%. These transmittance variations are associated with the corresponding frequencies of 1.005 THz, 1.004 THz, 1.002 THz, 1.000 THz, and 0.999 THz, respectively. These changes indicate the sensor's sensitivity to the target analyte (COVID-19), where the decrease in transmittance correlates with the binding of COVID-19-specific molecules to the sensor surface. Furthermore, a magnified view provided in

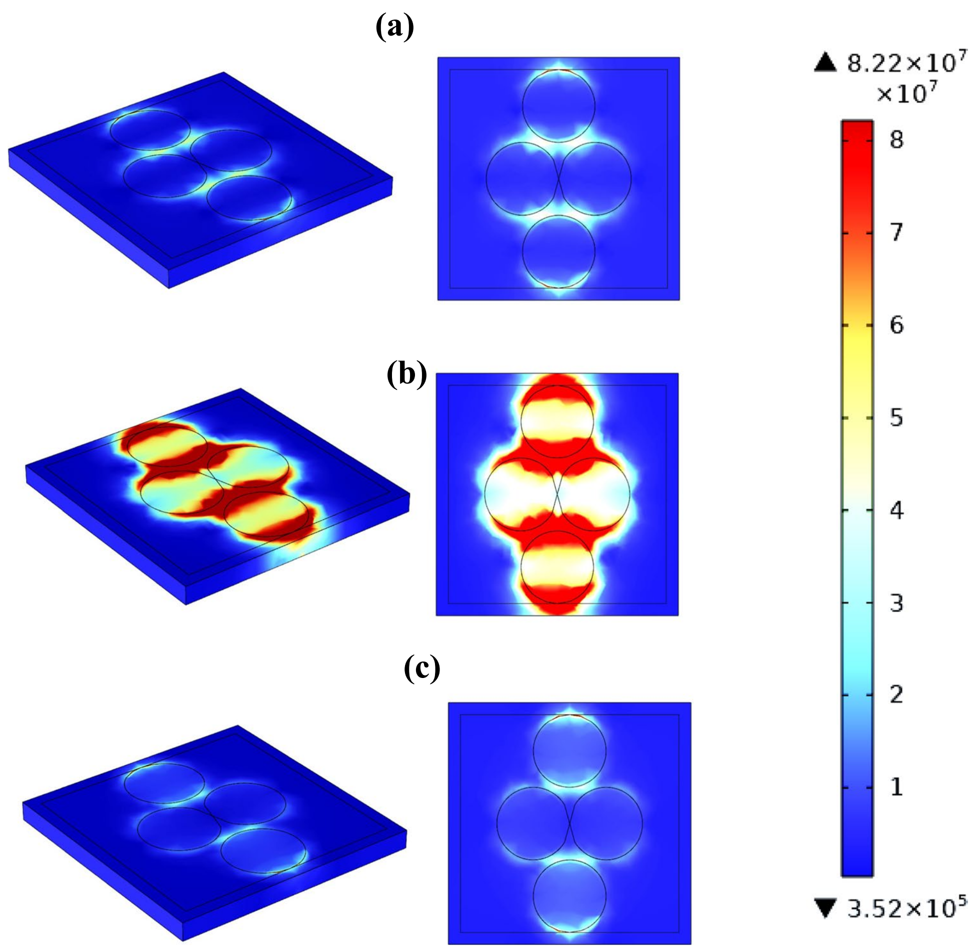

Fig. 7 Distribution of electric field intensity across the proposed sensor structure at three key frequencies: **a** 0.911THz, **b** 1.007 THz, and (**c**) 1.272THz

Table 2 Key performance metrics of the sensor design under investigation

f(THz)	1.005	1.004	1.002	1	0.999
n(RIU)	1.334	1.34	1.345	1.35	1.355
df(THz)		0.001	0.002	0.002	0.001
dn(RIU)		0.006	0.005	0.005	0.005
S(GHz/RIU)		167	400	400	200
FWHM(THz)	0.143	0.143	0.143	0.143	0.143
FOM(RIU^{-1})		1.166	2.797	2.797	1.399
Q	7.028	7.021	7.007	6.993	6.986
DL		1.978	0.693	0.693	1.648
DR	2.658	2.655	2.650	2.644	2.642
SNR		0.007	0.014	0.014	0.007
SR		0.330	0.277	0.277	0.330
DA	6.993	6.993	6.993	6.993	6.993
X		0.001	0.001	0.001	0.001

Fig. 5b reveals a detailed frequency shift range, which shows a shift from 1.03 THz to 0.97 THz. This shift corresponds to a tuning of 60 GHz, which is significant for accurately detecting the presence of COVID-19 and can be attributed to the specific interactions of the virus particles with the sensing elements. The shift in frequency is an important indicator of the sensor's sensitivity and ability to distinguish COVID-19 detection at different concentrations.

Figure 6a and b show the relationships between resonance frequency, refractive indices (RIs), and Covid-19 sample concentrations. The analysis reveals a linear correlation between resonance frequency and refractive index, described by the equation:

$$F = -0.3065RI + 1.4142 \tag{14}$$

Table 3 Comparison of the proposed sensor with other cases

	f	Quality factor	(S)	Materials	Application
[58]	0–6000 nm	-	6227.6 nm/RIU	gold	Temperature sensing
[59]	0.1–2.5THz	11.247	1149 GHz/RIU	Graphene	Metal ions
[60]	3.495 −3.945	-	1.41THz/RIU	metamaterials	Gas detection
[61]	0.5–1.4THz	5	300 GHz/RIU	Graphene	Biochemical application
[62]	-	-	4800 nm/RIU	Gold, α-iron (III) oxide	Biosensing
[63]	0.1-2THz	13.528	4318 GHz/RIU	Graphene	Organic Compounds
[64]	-	23.38597	163.71°/RIU	-	Helium detection
[65]	0.5–1.1THz	8.947	2000 GHz/RIU	Graphene	Pregnancy detection
Proposed sensor	0.6–1.4THz	7.028	400GHzRIU^{-1}	Graphene,	Detection of Covid-19

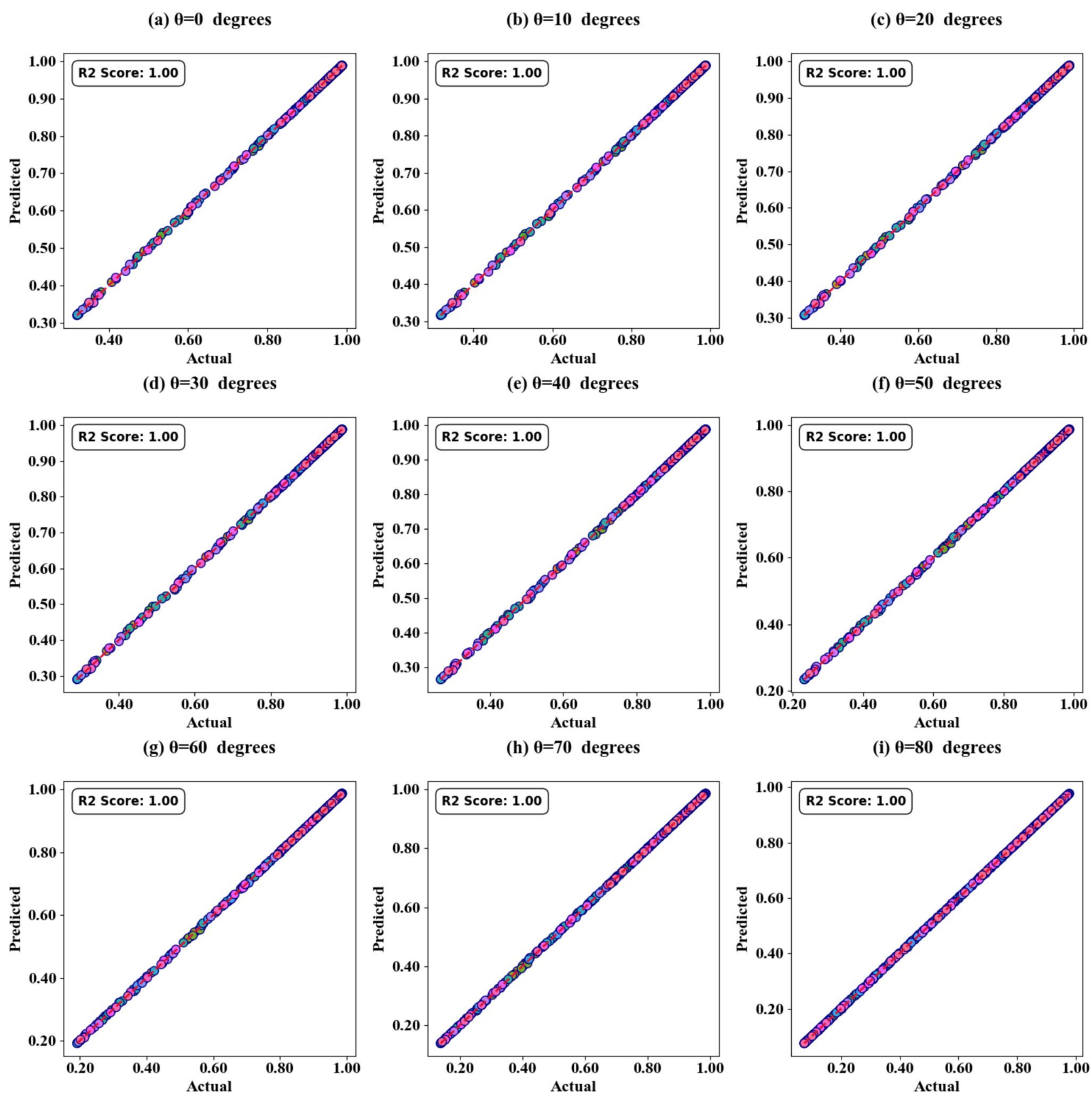

Fig. 8 presents scatter plots illustrating the relationship between actual and predicted absorption values for different Incident angle values

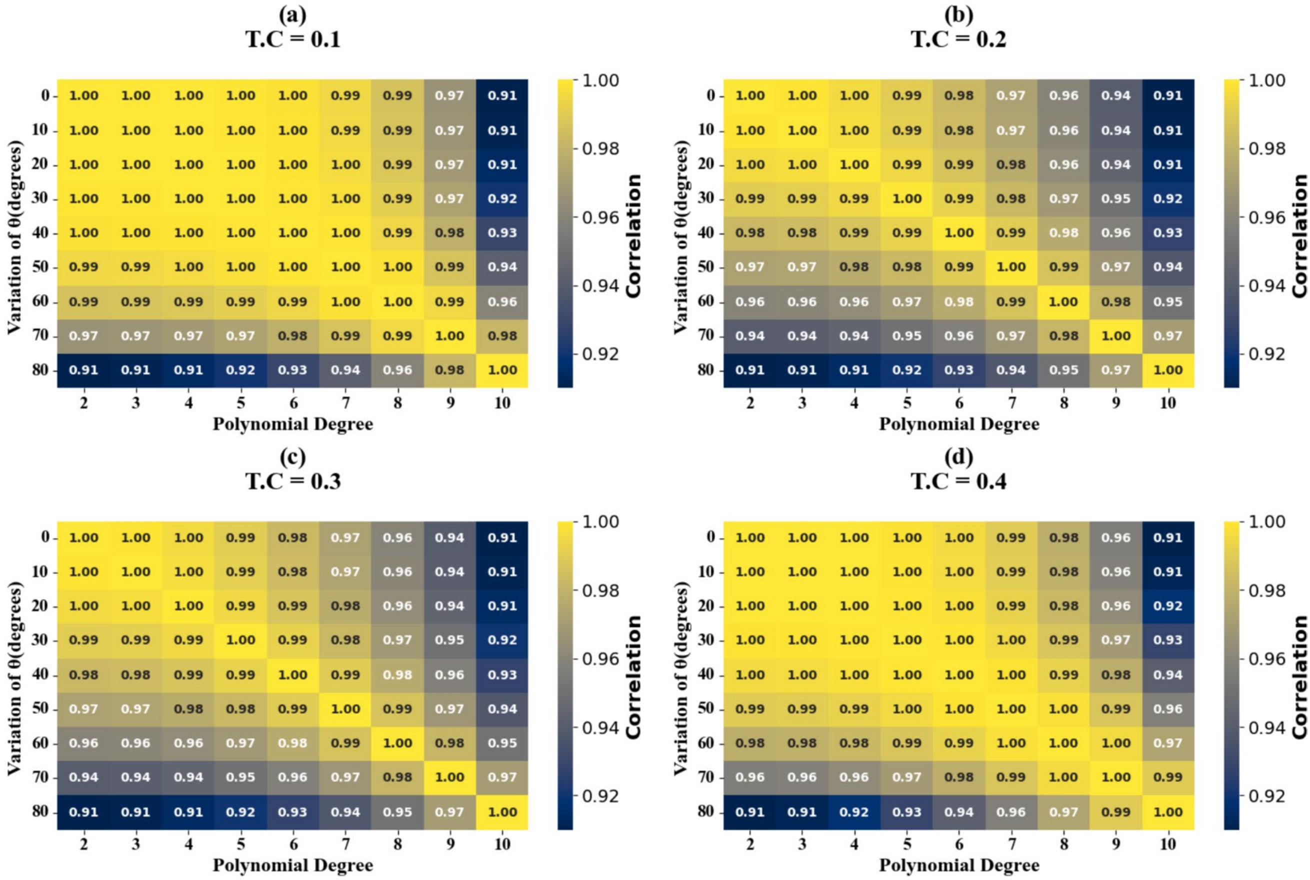

Fig. 9 presents Heat map plots illustrating the relationship between actual and predicted absorption values for different Incident angle values

This relationship has a coefficient of determination (R^2) of 97.844%. Similarly, resonance frequency and concentration are linearly related through the equation:

$$F = -KC + 1.0053 \tag{15}$$

This relationship also has an R^2 value of 97.844%. In this context, C represents concentration (Mm) and K is a constant.

Figure 7a-c display the electric field distributions within the sensor at three distinct frequencies: (a) 0.911 THz, (b) 1.007 THz, and (c) 1.272 THz, illustrating the interaction of electromagnetic waves with the sensor. At 0.911 THz (Fig. 7a) and 1.272 THz (Fig. 7c), the waves predominantly pass through the sensor with minimal interaction, as indicated by the deep blue regions in the field distributions, suggesting weak coupling and low energy absorption. In contrast, at 1.007 THz (Fig. 7b), the sensor exhibits notable resonance behaviour, with concentrated field regions (brown-coloured) that indicate strong coupling, increased field intensities, and substantial energy absorption.

The terahertz-range sensing device exhibits significant performance variations across a narrow frequency band (0.999–1.005 THz), despite minimal refractive index variations (1.334–1.355 RIU) as demonstrated in Table 2. Most notably, sensitivity (S) demonstrates frequency-dependent behavior, peaking at 400 $GHzRIU^{-1}$ and 1.004 THz while declining to 167 $GHzRIU^{-1}$. This sensitivity pattern directly correlates with the Figure of Merit (FOM), which reaches its maximum value of 2.797 RIU^{-1} at the same optimal frequencies, and Detection Limit (DL), which achieves its minimum value of 0.693 at these frequencies. Several performance

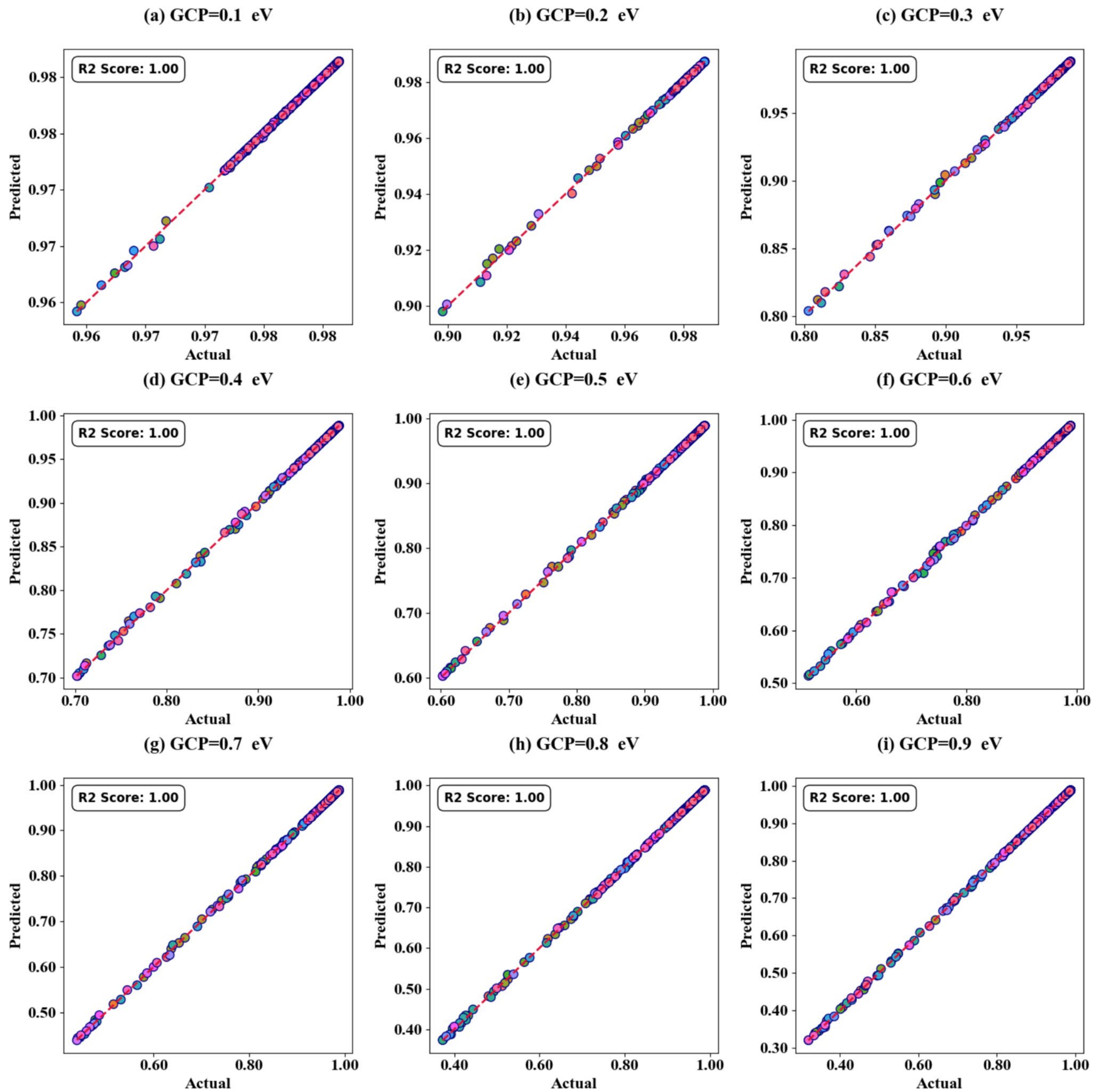

Fig. 10 presents scatter plots illustrating the relationship between actual and predicted absorption values for different GCP values

metrics remain consistent across the measured frequency range, including FWHM (0.143 THz), Detection accuracy (6.993), Quality Factor (approximately 7.0), and Dynamic Range (approximately 2.65). The Signal-to-Noise Ratio (SNR) varies between 0.007–0.014, with peak values corresponding to the optimal sensing frequencies. These findings suggest that the device demonstrates optimal sensing performance at 1.002 and 1.004 THz, exhibiting enhanced sensitivity and detection capability while maintaining structural resonance characteristics across the entire measured frequency range.

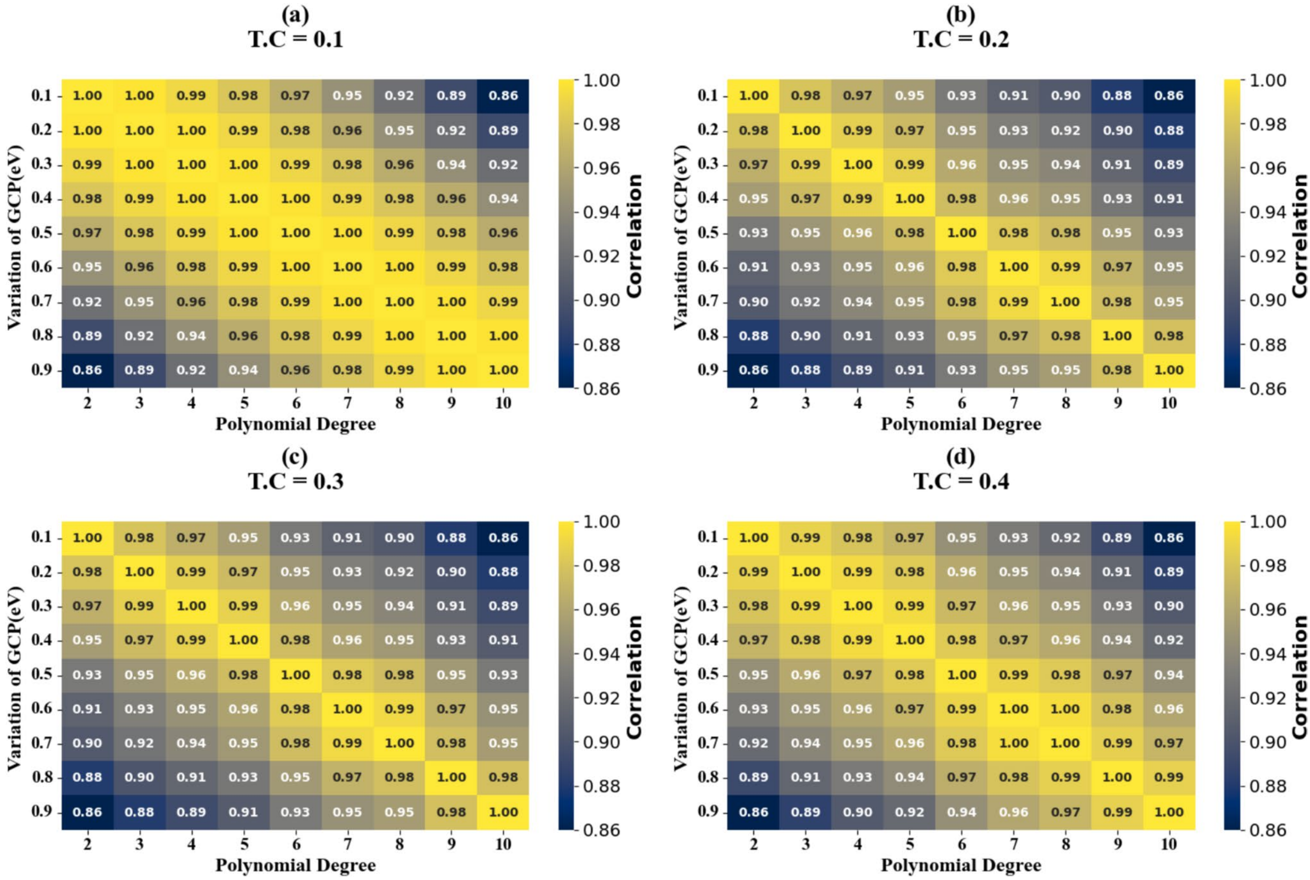

Fig. 11 presents Heat map plots illustrating the relationship between actual and predicted absorption values for different GCP values

The proposed sensor as demonstrated in Table 3 operates in the 0.6–1.4 THz range with a quality factor of 7.028 and a sensitivity of 400 GHz/RIU, using graphene for COVID-19 detection. It compares favorably to other sensors, which vary in frequency, sensitivity, and applications such as temperature sensing, metal ion detection, and biochemical applications. The sensor's performance is competitive with other graphene-based designs, particularly for specific applications.

Machine Learning Optimization Based on 1D Convolutional Neural Networks

Optimization techniques for 1D-CNNs involve adjusting hyperparameters such as learning rates and batch sizes, as well as using regularization methods like dropout and L_2 regularization to avoid overfitting[66]. Furthermore, methods like Bayesian optimization can be utilized to efficiently explore the best hyperparameter values, improving model performance and shortening training time. In general, 1D-CNNs offer a strong foundation for refining machine learning models in applications dealing with sequential or time-series data[67]

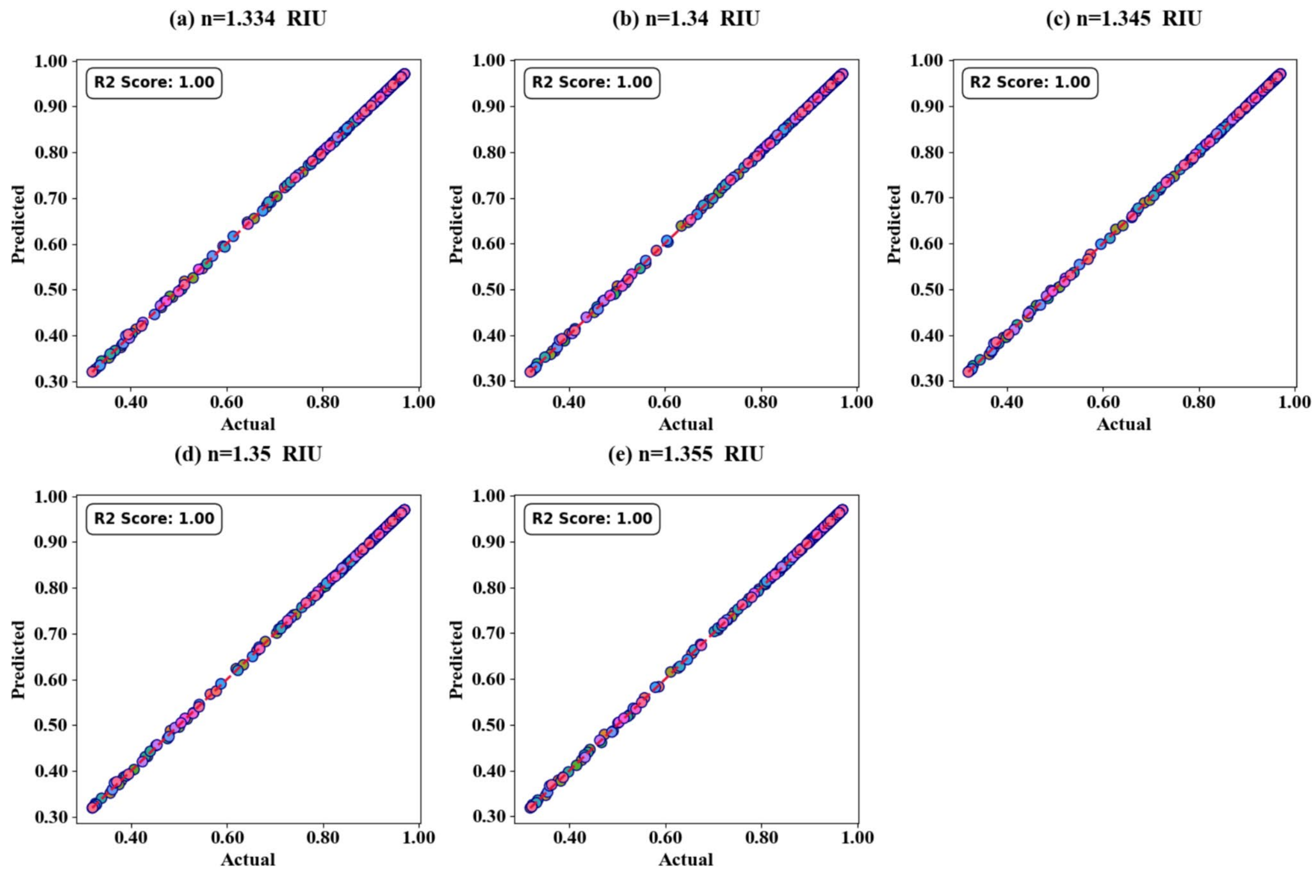

Fig. 12 presents scatter plots illustrating the relationship between actual and predicted absorption values for different RIs values

Some metrics used for analysing the model include[68];

$$(f \times g)(t) \tag{16}$$

$$(f \times g)(t) = \int f(\tau)g(t-\tau)d\tau \tag{17}$$

$$y = \frac{\gamma \times ((\mathrm{x} - \mu)}{\sqrt{(\sigma^2 + \varepsilon) + \beta}} \tag{18}$$

$$\mathrm{MSE} = (1/\mathrm{N})x\Sigma(\mathrm{i} = 1 \text{ to } \mathrm{N})\left(y_i - \hat{y}_n^2\right) \tag{19}$$

$$\mathrm{y} = \mathrm{f(x)} \times \mathrm{mask} \tag{20}$$

$$R^2 = \frac{\sum_{i=1}^{N}(PredictedTargetValue_i - ActualTargetValue_i)}{\sum_{i=1}^{N}\left(ActualTarget_i - AverageTargetValue\right)^2} \tag{21}$$

$$\theta = \theta - \alpha \times \nabla J(\theta) \tag{22}$$

The model's performance is evaluated using scatter plots and R^2 score heat maps to assess the combinations of polynomial degrees and incident angle parameters, as shown in Figs. 8a-i and 9a-d. The model achieves optimal performance with a perfect R^2 score of 100% at specific incident angle values, as demonstrated by both the scatter plots and heat maps.

The model's performance is assessed through scatter plots and R^2 score heat maps to evaluate the combinations of polynomial degrees and Graphene chemical potential

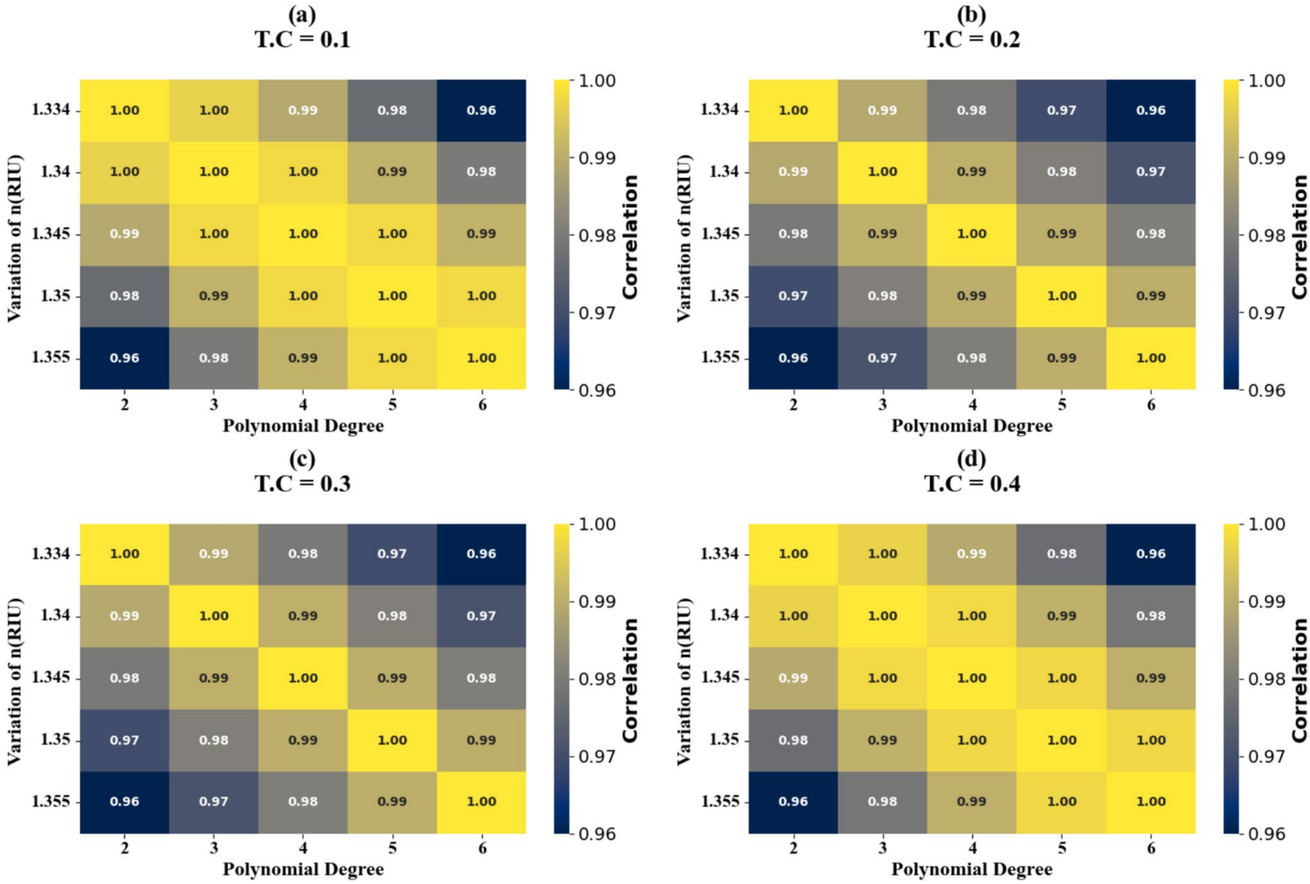

Fig. 13 presents Heat map plots illustrating the relationship between actual and predicted absorption values for different RIs values

(GCP) parameters, as illustrated in Figs. 10a-i and 11a-d. It achieves optimal performance with a perfect R^2 score of 100% at specific GCP values, as shown by both the scatter plots and heat maps.

The model's performance is evaluated using scatter plots and R^2 score heat maps to analyze the combinations of polynomial degrees and RI parameters, as depicted in Figs. 12a-e and 13a-d. It demonstrates optimal performance with a perfect R^2 score of 100% at specific RI values, as illustrated by both the scatter plots and heat maps.

Conclusion

This study demonstrates the successful development of a graphene metasurface-based SPR biosensor for COVID-19 detection, combining advanced materials engineering with machine learning optimization. The sensor's performance metrics, including its 400 GHzRIU^{-1} sensitivity and 7.028 quality factor, position it competitively among existing biosensing platforms. The integration of 1D-CNN optimization techniques significantly enhances the sensor's detection capabilities, achieving perfect R^2 scores under optimal conditions. The comprehensive analysis of various operational parameters. Future research directions could focus on further miniaturization, enhanced sensitivity through advanced materials integration, and expanded application to other viral pathogens.

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Declarations

Ethical Approval Not applicable.

Competing interests The authors declare no competing interests.

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